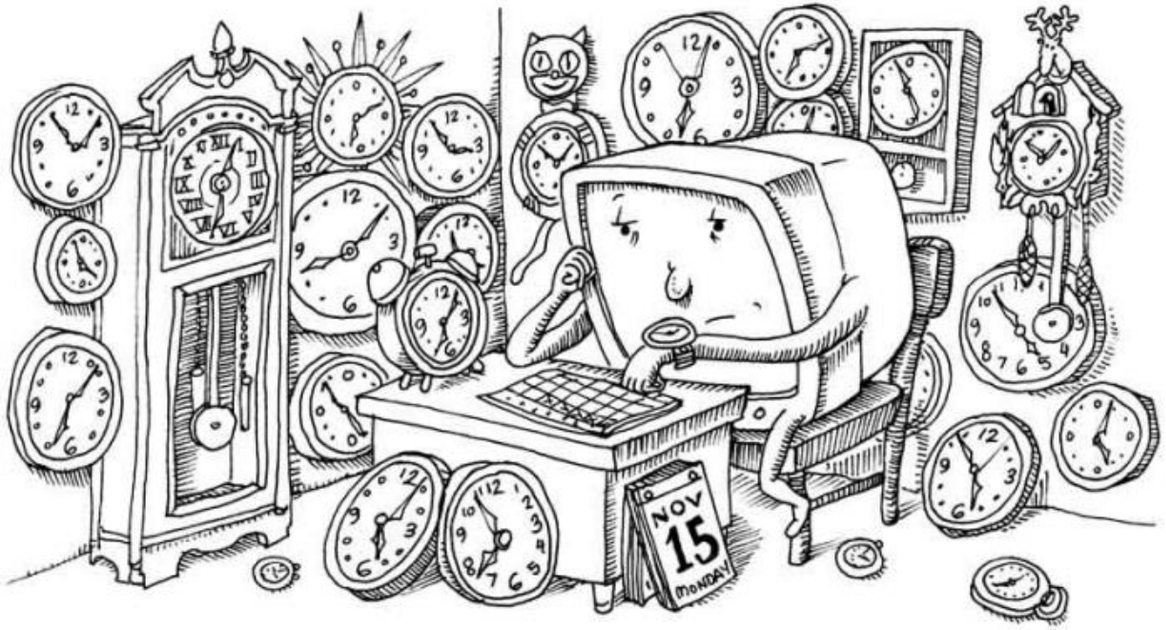


Periodic Processes

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Handbook and Manual pages

- FreeBSD
  - https://www.freebsd.org/doc/en/books/handbook/configtuning-cron.html (in English)
  - https://www.freebsd.org/doc/zh_TW/books/handbook/configtuning-cron.html (in Chinese)
- Debian
  - https://debian-handbook.info/browse/stable/sect.task-scheduling-cron-atd.html (in English)
  - https://debian-handbook.info/browse/zh-TW/stable/sect.task-scheduling-cron-atd.html (in Chinese)

CRON – Schedule Commands (1)

- What we want?
  - Do things at right time automatically
- cron daemon
  - The daemon that handles periodic execution
  - cron daemon reads configuration file and executes commands on time

cron(8), crontab(1), crontab(5)

CRON – Schedule Commands (2)

- Configuration file
  - So called: crontab (cron table)
  - Location of user cron configuration file
    - Every user can have at most one crontab file and this file will be named the user's login ID
    - Edit using crontab(1) command

System	Cron Dir
FreeBSD	/var/cron/tabs
Debian	/var/spool/cron/crontabs
Red Hat	/var/spool/cron

  - Location of System Cron Configuration file
    - /etc/crontab

CRON – Schedule Commands (3)

- Configuration File Format
  - Ignored
    - Blank lines or leading spaces and tabs
  - Comments
    - pound-sign lines whose first non-space character is a #
  - environment setting
    - name = value
    - Default environment variables
      - LOGNAME, SHELL, PATH, HOME, MAILTO

```
SHELL=/bin/sh
PATH=/etc:/bin:/sbin:/usr/bin:/usr/sbin:/usr/local/bin
```

  - cron command example:

```
# minute hour day month weekday command
  33     7    *   *     *       /usr/local/bin/rsync -al -delete /home/ backup:/raid/home/
```

CRON – Schedule Commands (4)

- cron command format – *minute hour day month weekday command*

Field	Description	Range
minute	Minute of the hour	0 ~ 59
hour	Hour of the day	0 ~ 23
day	Day of the month	1 ~ 31
month	Month of the year	1 ~ 12
weekday	Day of the week	0 ~ 6 (0 = Sunday)

- Rule Matching
  - * matches everything
  - Single character matches exactly
  - Dash(-) matches range
  - Comma(,) matches any listed value
  - Slash(/) matches skips of the number's value through the range.

CRON – Schedule Commands (5)

- crontab time format example

45 10 * * 1-5 → AM 10:45, from Mon. to Fri.
10 * * * * → On the 10th minute of each hour
*/3 * * * * → Every three minutes
30 15 5 * * → PM 3:30 of each 5th day
0 0 14 2 * → On the Midnight of Valentine's day
5 0-6 * * * → On the 5th minute, from 0 to 6 o'clock
0,30 * 13 * 5 → Every half-hour on Fri. and the 13th day

periodic e.g., 1-59/2 = 1, 3, 5, 7, 9, …, 59

- crontab example

```
20  1  *  *  *          find /tmp -atime +3 -exec rm -f {} ‘;’
55  23  *  *  0-3,6     /home/tsaimh/cputemp-check.sh
```

CRON – Schedule Commands (6)

- Special strings to specify the time

string	meaning	in 5 fields format
@reboot	Run once, at startup.	N/A
@yearly	Run once a year	0 0 1 1 *
@annually	(same as @yearly)	
@monthly	Run once a month	0 0 1 * *
@weekly	Run once a week	0 0 * * 0
@daily	Run once a day	0 0 * * *
@midnight	(same as @daily)	
@hourly	Run once an hour	0 * * * *
@every_minute	Run once a minute	*/1 * * * *
@every_second	Run once a second	

crontab command

- crontab(1)

$ crontab -e [-u user]

- Edit the [user's] crontab using editor

$ crontab -l

- List the content of the crontab

$ crontab -r

- Remove the current crontab

$ crontab *filename*

- Install *filename* as your crontab

crontab management (1)

- To Allow or deny user from using cron daemon
  - By default, all users can have their own crontab
  - allow file
    - A list of users that may use crontab, any other not in the list can not use it
  - deny file
    - Reverse meaning
- log

System	Allow or deny file	Log file
FreeBSD	/var/cron/{allow,deny}	By syslogd
Debian / Red Hat	/etc/cron.{allow,deny}	/var/log/cron

System crontab: /etc/crontab

- In FreeBSD

```
SHELL=/bin/sh
PATH=/etc:/bin:/sbin:/usr/bin:/usr/sbin
HOME=/var/log
#minute hour   mday   month wday   who      command
*/5     *      *      *     *      root     /usr/libexec/atrun
*/11    *      *      *     *      operator /usr/libexec/save-entropy
0       *      *      *     *      root     newsyslog
1       3      *      *     *      root     periodic daily
15      4      *      *     6      root     periodic weekly
30      5      1      *     *      root     periodic monthly
1,31    0-5    *      *     *      root     adjkerntz -a
```

1

2

System crontab: /etc/crontab (cont.)

- In Linux

```
# .---------------- minute (0 - 59)
# |  .------------- hour (0 - 23)
# |  |  .---------- day of month (1 - 31)
# |  |  |  .------- month (1 - 12) OR jan,feb,mar,apr ...
# |  |  |  |  .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# |  |  |  |  |
# *  *  *  *  * user-name command to be executed
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.daily; }
47 6    * * 7   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.weekly; }
52 6    1 * *   root    test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.monthly; }
```

anacron, usually used for laptops, performs periodic command scheduling without assuming that the system is running continuously. You may install it if needed.

periodic utility (1)

- periodic(8)
  - Run periodic system function under /etc/periodic

```
$ ls -ld /etc/periodic/*
drwxr-xr-x  2 root  wheel  1024 Sep 26 21:43 /etc/periodic/daily
drwxr-xr-x  2 root  wheel   512 Sep 27 03:49 /etc/periodic/monthly
drwxr-xr-x  2 root  wheel   512 Sep 27 03:49 /etc/periodic/security
drwxr-xr-x  2 root  wheel   512 Sep 27 03:49 /etc/periodic/weekly
```

```
$ ls /etc/periodic/daily
100.clean-disks      200.backup-passwd    405.status-ata-raid  430.status-rwho
110.clean-tmps       210.backup-aliases   406.status-gmirror   440.status-mailq
120.clean-preserve   300.calendar         407.status-graid3    450.status-security
130.clean-msgs       310.accounting       408.status-gstripe   470.status-named
140.clean-rwho       330.news             409.status-gconcat   500.queuerun
150.clean-hoststat   400.status-disks     420.status-network   999.local
```

- periodic utility
  - /etc/periodic.conf
  - /etc/defaults/periodic.conf

```
$ ls -al /usr/local/etc/periodic/security
total 18
drwxr-xr-x  2 root  wheel      4 Apr 12  2017 .
drwxr-xr-x  8 root  wheel      8 Aug 20  2016 ..
-r-xr-xr-x  1 root  wheel  4944 Apr  2  2017 410.pkg-audit
-r-xr-xr-x  1 root  wheel  1686 Apr  2  2017 460.pkg-checksum
```

periodic utility (2)

- For custom system programs: /usr/local/etc/periodic

```
$ ls -l /usr/local/etc/periodic/
total 19
drwxr-xr-x  2 root  wheel  6 Apr 12  2017 daily
drwxr-xr-x  2 root  wheel  4 Aug 20  2016 hourly
drwxr-xr-x  2 root  wheel  4 Aug 20  2016 monthly
drwxr-xr-x  2 root  wheel  3 Aug 20  2016 reboot
drwxr-xr-x  2 root  wheel  4 Apr 12  2017 security
drwxr-xr-x  2 root  wheel  6 Apr 12  2017 weekly
```

```
$ ls -l /usr/local/etc/periodic/daily/
total 18
-r-xr-xr-x  1 root  wheel  1512 Jul 29  2016 402.zfSnap
-r-xr-xr-x  1 root  wheel  1073 Jul 29  2016 403.zfSnap_delete
-r-xr-xr-x  1 root  wheel  2746 Apr  2  2017 411.pkg-backup
-r-xr-xr-x  1 root  wheel  2506 Apr  2  2017 490.status-pkg-changes
```



periodic utility (3)

- Execution order depends on filenames
  - Use number as prefix to control the order
- All scripts under that directory will be executed
  - Unlike /etc/rc.conf
  - Even though there is no "YES" in /etc/periodic.conf
- /etc/periodic.conf

```
$ cat /etc/periodic.conf
daily_clean_tmps_enable="YES"                   # Delete stuff daily
daily_clean_tmps_dirs="/tmp /var/tmp"           # Delete under here
daily_clean_tmps_days="3"                       # If not accessed for
daily_status_zfs_enable="YES"                   # Check ZFS
daily_status_ntpd_enable="YES"                  # Check NTP status
weekly_dehydrated_enable="YES"
...
```

- Driven by crontab(1)

at : Deferred execution scheduler

- at(1)
  - executes commands at a specified time

    at [-q queue] [-f file] [-mldbv] time

    or at [-q queue] [-f file] [-mldbv] -t [[CC]YY]MMDDhhmm[.SS]
- at management
  - atq: View job queue
  - atrm: Remove jobs
  - /var/at/at.{allow,deny} (FreeBSD)
  - /etc/at.{allow,deny} (Linux)
    - In FreeBSD, by default, only root can execute "at" command
    - In Linux, by default, every user can execute "at" command

at : Deferred execution scheduler (cont.)

```
tsaimh@linux1:~$ at -t 10270901
warning: commands will be executed using /bin/sh
at Mon Oct 27 09:01:00 2025
at> echo 0901 > ~/at_file1
at> echo hihi >> ~/at_file1
at> <EOT>
job 22 at Mon Oct 27 09:01:00 2025
tsaimh@linux1:~$ atq
22      Mon Oct 27 09:01:00 2025 a tsaimh
tsaimh@linux1:~$ at -t 10270903
warning: commands will be executed using /bin/sh
at Mon Oct 27 09:03:00 2025
at> echo 0903 > ~/at_file2
at> echo hehe
at> <EOT>
job 23 at Mon Oct 27 09:03:00 2025
```

```
tsaimh@linux1:~$ atq
23      Mon Oct 27 09:03:00 2025 a tsaimh
tsaimh@linux1:~$ date
Mon Oct 27 09:01:25 AM CST 2025
tsaimh@linux1:~$ cat ~/at_file1
0901
hihi
tsaimh@linux1:~$ date
Mon Oct 27 09:03:51 AM CST 2025
tsaimh@linux1:~$ cat ~/at_file2
0903
tsaimh@linux1:~$ atrm 22
Cannot find jobid 22
```

at : Deferred execution scheduler (cont.)

- In FreeBSD, at is driven by crontab(1).
  - Invoked every 5 minutes

```
*/5     *       *       *       *       root    /usr/libexec/atrun
```

- In Linux, at is driven by atd.service
  - Ensure your atd is active by systemctl status atd

```
$ systemctl status atd
● atd.service - Deferred execution scheduler
     Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled;
     Active: active (running) since Tue 2025-09-23 06:53:47 CST; 1 month
     Docs: man:atd(8)
     …
```